

ELI — Memory

ELI v2.4.68

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Contents

ELI Memory Subsystem	1
The two layers	1
Files	2
The storage policy (policy.py)	2
The Memory class (memory.py)	3
Schema	3
Shared turn retrieval (retrieval.py)	4
recall_memory: the hybrid retriever	4
Vector store (vector_store.py)	4
Knowledge graph (knowledge_graph.py)	5
Truth layer (memory_truth.py)	5
Promotion across tiers	5
Honest assessment	5
Recent-history window	5

ELI Memory Subsystem

Updated for v2.4.68. Memory is governed by a storage policy (eli/memory/policy.py): every row has an origin, repeats are merged, weight follows a forgetting curve reinforced by use, and faded derived rows are archived. Recall applies a time window before ranking, and reports what it did. Turn retrieval is shared in eli/memory/retrieval.py; FAISS deletes use tombstones.

eli/memory/ holds 8,770 lines in 11 files: relational, full-text, vector and graph storage, all local SQLite and FAISS. Companion to project_overview.md.

The two layers

ELI’s memory is not “a dump of every interaction”. There are two layers:

1. **Conversation history** (conversation_turns, conversations, session_summaries): broad and chronological. A turn is kept unless the persistence gate (runtime/persistence_gate.py) rejects it as junk. This is what a question about a period of time reads.
2. **Distilled memory** (memories, semantic, user_patterns, the knowledge graph): a smaller set that the gate and the storage policy judged worth keeping, labelled by origin, merged when repeated, and faded when nothing reinforces it. Faded derived rows move to memories_archive instead of being deleted.

The memory-runtime report states this explicitly (EXPLAIN_MEMORY_RUNTIME, memory internals), with live counts for each layer.

Files

File	Lines	Role
memory.py	5,677	the Memory class (69 public methods), schema, DBPaths, upkeep, module facade
policy.py	118	pure storage-policy functions (origin, dedupe key, strength, archive, merge)
retrieval.py	186	shared turn retrieval (retrieve_for_turn), turn cache, time window
unified_retrieval.py	168	the orchestrator stages consume retrieve_for_turn through it; formats the verified-memory block
vector_store.py	637	FAISS index, embedder, tombstones
knowledge_graph.py	643	entity and relation graph
habits_memory_db.py	466	habit and legacy memory helpers
system_index.py	278	indexed apps, executables, files
memory_truth.py	188	read-only inspection used by status surfaces
memory_adapter.py	131	compatibility adapter
__init__.py	278	facade (get_memory, get_agent_memory, path helpers)

The storage policy (policy.py)

Every stored row gets an **origin** from `classify_origin(source, kind, tags, text)`:

Origin	Meaning	Factor on half-life
user_said	the owner said it or confirmed it	1.0
eli_said	ELI's own assistant text	0.5
tool	executor or observation output	0.35
news	news digests and reflections	0.25
telemetry	bookkeeping (<code>is_bookkeeping_memory</code>)	0.15

Rules, all in `policy.py` and pinned by `tests/test_memory_policy.py`:

- **Dedupe.** `text_key` is a sha1 of the normalised text (first 20 hex characters). A repeat bumps `seen_count` and `last_seen` instead of adding a row; `merge_group` keeps the earliest event time.
- **Strength.** $2^{**} (-days_since_last_touch / half_life)$, floored at `MIN_WEIGHT` (0.05). The half-life is the base half-life $\times$ origin factor $\times (1 + 3 \times importance) \times (1 + 0.6 \times \ln(1 + uses))$, where `uses` are repeat sightings plus recalls.
- **Base half-life.** 30 days, scaled by the square root of the median gap between days the owner used ELI, clamped to 14–120 days (`adaptive_half_life_days`).
- **Pinning.** A `user_said` row with `importance` ≥ 0.85 stays at strength 1.0.
- **Archiving.** `should_archive` is true only for a derived (not `user_said`) row that has never been recalled, has `importance` < 0.85 , and whose `weight` is ≤ 0.12 .

- **Vector index.** Telemetry rows are not embedded (wants_vector_index).
- **Evidence about the user.** Only user_said counts (counts_as_evidence_about_user).

Columns added to memories: event_ts, seen_count, last_seen, last_recalled, recall_count, origin, text_key. Tables added: memories_archive, memory_meta. The semantic table gained evidence_count, last_seen, evidence.

Daily upkeep

Memory.run_upkeep() runs at most once a day (upkeep_due()), started asynchronously at engine start and after responses (upkeep_async()), and skipped under ELI_TEST_MODE. Steps:

1. **backfill:** label existing rows with origin, key and event time (after copying the database to a .pre_policy.bak file the first time);
2. **consolidate:** merge exact duplicates and rebuild memories_fts;
3. **decay:** recompute weight with the policy strength (idempotent: a pure function of age, importance and use, so a missed run cannot leave a row over-weighted);
4. **archive:** move faded derived rows to memories_archive (search_archive, restore_from_archive bring them back);
5. **identity:** merge legacy user ids into one stable owner id (owner_id, kept in memory_meta); only UUID-shaped legacy ids are merged;
6. **learning:** remove repeated identical belief revisions, cap corroboration at the number of days the owner has used ELI, and drop ledger copies of replay rows;
7. **semantic:** fold repeated facts into one row with evidence_count and up to five quotes.

integrity_report() reports the state of all of this for the health checks.

The Memory class (memory.py)

A single class that owns most persistent concerns:

- **Memory:** store_memory, add_memory, recall_memory, search_memory, search_memories, memories_between, get_recent_memories, adjust_weight, apply_weight_decay, consolidate_memories, archive_faded, search_archive, restore_from_archive, store_episodic, store_semantic, recall_semantic, store_reflective.
- **Conversation:** add_conversation_turn, store_conversation, get_conversation_history, get_recent_conversations (with since, until, role), get_recent_turns_since, search_conversations, get_turns_for_day, save_session_summary, get_session_summaries.
- **Habits:** log_habit_event, get_habit_events, add_habit_rule, get_habit_rules, get_detected_habits, record_habit_run.
- **Self-improvement and learning:** log_learning_event, log_failure, log_correction, add_observation, log_improvement, add_capability_proposal, propose_capability, get_pending_proposals, get_recent_failures, get_recent_improvements, get_recent_observations. These tables are written on the **agent** database (SelfImprovementEngine opens get_agent_memory()); the user database holds empty copies.
- **Health:** get_stats, get_dashboard_counts, get_db_routing_info, integrity_report.

vector_store is a lazy property; the knowledge graph is reached through knowledge_graph.get_knowledge_graph().

Schema

The blank user.sqlite3 template holds 27 tables and three FTS5 indexes. memories (+ memories_fts), memories_archive, memory_meta, semantic, conversation_turns, conversations, session_summaries, kg_entities (+ FTS), kg_relations, recall_log, runtime_events, learning_replay, observations, habits, habit_events, habit_rules, user_patterns, user_model, eli_stances, belief_revisions, corrections, failures, error_tracking, improvements, capability_proposals, news_articles (+ FTS), news_reflections. agent.sqlite3 holds agent_dispatches, agent_metrics and the self-improvement tables.

emotion_events is owned by cognition/emotion_timeline.py, not the Memory class: it opens user.sqlite3 directly and creates its table on first use. Columns: ts, user_id, session_id, detected, expressed, valence (negative, positive or neutral), confidence, source, arousal, user_text and eli_prior_action (the action ELI ran on the preceding turn, so it can ask whether the mood turned because of something it did). Reads come back ORDER BY ts DESC, id DESC; the id tiebreak matters because several reads can land within one second.

Each conversation turn is stored in conversation_turns. A learning_replay row is also written per turn (it carries action, outcome and reward), and the evidence ledger records the turn once in runtime_events; replay rows are not mirrored into the ledger.

Shared turn retrieval (retrieval.py)

The single owner of semantic and conversation recall for a turn. BusMemoryAgent and the orchestrator both call retrieve_for_turn(), so a query is not searched twice with different budgets.

- **Turn cache:** 8 seconds per process. The key includes session, user, query, time window and every limit (semantic_limit, conv_limit, recent_limit, summary_limit, hop2_limit, merge_cap, verified_only), so a deeper pass never gets the shallower result. invalidate_turn_cache() clears it.
- **Time window first.** When the planner finds a period in the question (query_planner.parse_window), memories dated inside it are fetched by date (Memory.memories_between, ordered by importance × weight) and merged with the topic hits, hits outside the window are dropped, and the user's turns inside the window are read from conversation_turns. The result carries window_stats (candidates, added by date, in window, turns) and the orchestrator logs and reports it (memory_diag).
- **Hop-2 deepening** when the first hits are sparse; **heuristic rerank** via rerank_candidates(); **contradiction detection** through the bus.
- **Dated evidence.** Each recalled line carries its event date (evidence_format); working memory keeps the original date rather than the date it was re-pinned.

recall_memory: the hybrid retriever

The primitive retrieve_for_turn() builds on:

1. **FAISS and FTS5 both run** and are fused by reciprocal rank (fuse_ranked_lists); a LIKE scan covers the case where FTS returns nothing. keyword_only=True skips FAISS.
2. **Noise filtering** (memory_exclusion_sql in core/self_provenance.py): ELI's own bookkeeping kinds and sources never resurface as recalled user memories.
3. **Importance-weighted ordering** via COALESCE(importance, 0.5).
4. **Reinforcement.** A recalled row gets last_recalled and recall_count updated and its weight reset to 1.0; the recall is written to recall_log with its memory_id.

Column detection (_memory_table_columns) guards against schema drift between versions.

Vector store (vector_store.py)

FAISS IndexFlat, embeddings from a local nomic embedder (llama.cpp).

- _embed_lock (RLock) serialises embedding: the embedder is not thread-safe, which is why orchestrator retrieval is sequential.
- Metadata lives in meta.json (migrated from a legacy meta.pkl).
- Singleton via get_vector_store(); shutdown-aware; reset_vector_store() for rebuilds.
- **Tombstones:** mark_memory_deleted(id) writes to meta.tombstones.json next to the metadata; search skips tombstoned rows without a rebuild; compact_tombstones(live_ids) reclaims space.
- store_memory() returns vector_indexed: bool, a failed index write logs at warning level, and fire_memory_uncertainty_event() drives the World tab's memory_uncertainty bar.
- Tombstoning is a per-call-site discipline: there is no shared delete_memory_row() helper, so any direct DELETE FROM memories must tombstone the vector itself (consolidate_memories and profile_extractor._scrub_onboarding_snapshot do).

Knowledge graph (knowledge_graph.py)

kg_entities(name, type, aliases, description, confidence) and kg_relations(subject_id, predicate, object_id, weight, source): a subject-predicate-object graph. FTS5 over entities with insert, update and delete triggers for fuzzy search_entities. upsert_entity; context_for_prompt gives lightweight SQLite-only prompt context with no embedding. A stop-word list stops common words becoming entities.

Truth layer (memory_truth.py)

inspect_sqlite and inspect_vector_store: read-only inspection used by status surfaces; reads vectors/meta.json and falls back to the legacy pickle. Backs truth_report and the memory-status surfaces.

Promotion across tiers

Promotion is a fan-out from user_patterns:

```
turns → user_patterns —→ semantic (profile_extractor._promote_to_semantic)
                        ↳ KG entities and relations
                          (persona_updater._populate_kg_from_user_patterns)
```

_promote_to_semantic applies a durable/transient prefix filter, dedupes against semantic and fires only when _insert_user_pattern actually inserted, so reaffirmations do not re-promote. Corroboration counts days the owner reaffirmed a fact, not extraction passes, and a passing mention cannot overwrite a fact the owner stated and reaffirmed (_insert_user_pattern weighs provenance and corroboration; each supersession is recorded in belief_revisions, and repeats are deduplicated). store_episodic and store_semantic are thin aliases and are not the promotion path.

Honest assessment

- **Strong:** genuinely hybrid (vector, FTS5, graph) on one local SQLite foundation; a measurable storage policy with an audit trail; recall that filters by date before ranking; noise filtering that keeps ELI's own reflections out of recall; embedder serialisation that avoids the segfault.
- **Weak:**
 1. memory.py is a 5.7k-line class spanning unrelated concerns (semantic, conversation, habits, learning, failures, capabilities, upkeep). It wants splitting along those seams.
 2. **Schema sprawl.** memories and a legacy memory, conversations and conversation_turns, plus a standalone semantic table; inline column detection compensates for schema churn.
 3. **Narrow intake.** Promotion works, but it is fed by regex extraction plus one LLM pass per session, so user_patterns, the semantic tier and the knowledge graph stay small relative to the conversation history. Widening the extractor is the work.
 4. **Retrieval cannot detect nonsense.** Similarity bands for genuine and gibberish queries overlap, so no absolute similarity floor separates them; the relative cutoff in vector_store tightens the candidate pool but is not a relevance oracle.
 5. **No shared delete-and-tombstone helper** (above).

Recent-history window

The user-facing tunable cog.mem_recent_turns (Settings ► Cognition; default 24, maximum 80) controls how many recent turns enter the prompt. RECENT_HISTORY_CAP no longer exists (the old memory-evidence module was removed); the prompt assembler and context_budget do the budgeting, and a recall question keeps a floor of a third of the window for memory context.